

The LaTeX Beamer `dmpisa` theme

Template with explanation of the main features

Bachelor, master, Ph.D., and Research

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This template is based on SINTEF Presentation from Federico Zenith and its derivation Beamer-LaTeX-Themes from Liu Qilong and Andrea Gasparini, and Mattia Ippoliti.

In the following you find a brief introduction on how to use LaTeX and the beamer package to prepare slides, based on the one written by Federico Zenith for SINTEF Presentation



Table of Contents

1 Introduction

Introduction

Personalization

Colours and Backgrounds

Summary



Beamer for `dmpisa` slides

1 Introduction

- We assume you can use LaTeX; if you cannot, you can learn it here
- Beamer is one of the most popular and powerful document classes for presentations in LaTeX
- Beamer has also a detailed user manual
- Here we will present only the most basic features to get you up to speed



Beamer vs. PowerPoint

Compared to PowerPoint, using LaTeX is better because:

- It is not What-You-See-Is-What-You-Get, but What-You-*Mean*-Is-What-You-Get: you write the content, the computer does the typesetting
- Produces a `pdf`: no problems with fonts, formulas, program versions
- Easier to keep consistent style, fonts, highlighting, etc.
- Math typesetting in TeX is the best:

$$\mathrm{i}\,\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = -\frac{\hbar^2}{2\,m}\nabla^2\Psi(\mathbf{r},t) + V(\mathbf{r})\Psi(\mathbf{r},t)$$



Getting Started

Selecting the dmpisa Theme

To start working with dmpisa, start a LaTeX document with the preamble:

Minimum dmpisa Beamer Document

```
\documentclass{beamer}
\usetheme{dmpisa}
\begin{document}
\begin{frame}{Hello, world!}
\end{frame}
\end{document}
```



Title page

1 Introduction

To set a typical title page, you call some commands in the preamble:

The Commands for the Title Page

```
\title{Sample Title}
\subtitle{Sample subtitle}
\author{First Author, Second Author}
\date{\today} % Can also be (ab)used for conference name &c.
```

You can then write out the title page with `\maketitle`.
To set a **background image** use the `\titlebackground` command before `\maketitle`; its only argument is the name (or path) of a graphic file.
If you use the **starred version** `\titlebackground*`, the image will be clipped to a split view on the right side of the title slide.



Writing a Simple Slide

It's really easy!

- A typical slide has bulleted lists



Writing a Simple Slide

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- A typical slide has bulleted lists
- These can be uncovered in sequence



Writing a Simple Slide

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Code for a Page with an Itemised List

```
\begin{frame}{Writing a Simple Slide}
  \framesubtitle{It's really easy!}
  \begin{itemize}[<+->]
    \item A typical slide has bulleted lists
    \item These can be uncovered in sequence
  \end{itemize}\end{frame}
```



Table of Contents

2 Personalization

Introduction

Personalization

Colours and Backgrounds

Summary



Language

By default, some text such as “Theorem”, “Definition”, as well as the format of the date, is in English. To change to your language (say, Italian), you must add `[italian]` in the document class, and use the package `babel`, as follows:

```
\documentclass[italian]{beamer}
\usepackage{babel}
```

which will result in the following change

Theorem (Weierstrass)

The header of this theorem is in English

Teorema (Weierstrass)

The header of this theorem is in Italian

Note that the “Table of Contents” won’t be translated by this since it is hardcoded in the style file, and must be manually modified



Theorem appearance

2 Personalization

By default, theorems and definitions appear as follows:

Theorem (Weierstrass)

if f is continuous on $[a, b]$, then f must attain a maximum and a minimum, each at least once

You can compress them and make them look like this by adding `[headerlesstheorems]` in the `usetheme` as shown below

Theorem 2 (Weierstrass): if f is continuous on $[a, b]$, then f must attain a maximum and a minimum, each at least once

```
\usetheme[headerlesstheorems]{dmpisa}
```



Changing Slide Style

- You can select the white or *maincolor* **slide style** *in the preamble* with `\themecolor{white}` (default) or `\themecolor{main}`
  - You should *not* change these within the document: Beamer does not like it
  - If you *really* must, you may have to add `\usebeamercolor[fg]{normal text}` in the slide
- You can change the **footline colour** with `\footlinecolor{color}`
  - Place the command *before* a new `frame`
  - There are four "official" colors: `maincolor`, `sintefyellow`, `sintefgreen`, `sintefdarkgreen`
  - Default is no footline; you can restore it with `\footlinecolor{}`
  - Others may work, but no guarantees!



The Footline

Options and defaults

- By default the footline is transparent and shows only the frame number
- `\footlinecolor{color}` colours it and adds the payoff (`author | title`)
- `\footlinecolor{}` makes it transparent again and removes the payoff
- `\footlinepayoff*` restores the default payoff on a transparent footline; `\footlinepayoff{text}` replaces it with custom text
- `\footlinesubtitletrue` appends the document subtitle to the default payoff (or enable it globally with `\usetheme[footlinesubtitle]{dmpisa}`)

Code for This Footline

```
\footlinecolor{}
\footlinepayoff*
\footlinesubtitletrue
```

Standard Blocks

These have a color coordinated with the footline (and grey in the blue theme)

```
\begin{block}{title}
content...
\end{block}
```

Colour Blocks

Similar to the ones on the left, but you pick the colour. Text will be white by default, but you may set it with an optional argument.

```
\begin{colorblock}[black]{sinteflightgreen}{title}
content...
\end{colorblock}
```

The “official” colours of colour blocks are: `sinteflilla`, `maincolor`, `sintefdarkgreen`, and `sintefyellow`.



Using Colours

2 Personalization

- You can use colours with the `\textcolor{<color name>}{text}` command
- The colours are defined in the `dmpisacolor` package:
  - Primary colours: `maincolor` and its sidekick `secondcolor`
  - Three shades of green: `sinteflightgreen`, `sintefgreen`, `sintefdarkgreen`
  - Additional colours: `sintefyellow`, `sintefred`, `sinteflilla`
    - These may be shaded—see the `sintefcolor` documentation or the SINTEF profile manual
- Do *not* abuse colours: `\emph{}` is usually enough
- Use `\alert{}` to bring the focus somewhere



Using Colours

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  - Additional colours: `sintefyellow`, `sintefred`, `sinteflilla`
    - These may be shaded—see the `sintefcolor` documentation or the SINTEF profile manual
- Do *not* abuse colours: `\emph{}` is usually enough
- Use `\alert{}` to bring the focus somewhere
- If you highlight too much, you don't highlight at all!

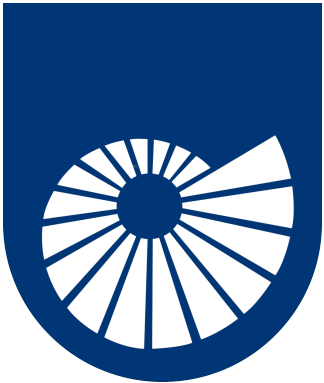


Adding images

Adding images works like in normal LaTeX:

Code for Adding Images

```
\usepackage{graphicx}
% ...
\includegraphics[width=\textwidth]
{assets/logo_RGB}
```





Splitting in Columns

Splitting the page is easy and common; typically, one side has a picture and the other text:

This is the first column

And this the second

Column Code

```
\begin{columns}
    \begin{column}{0.6\textwidth}
        This is the first column
    \end{column}
    \begin{column}{0.3\textwidth}
        And this the second
    \end{column}
    % There could be more!
\end{columns}
```

Special Slides

2 Personalization

- Chapter slides
- Side-picture slides

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Chapter slides

- Similar to `frames`, but with a few more options
- Chapter slides always use the style opposite to the document theme: dark in the white theme, light in the inverted theme
- An explicit colour keeps the dark treatment and colours the wordmark
- Opened with `\begin{chapter}[<image>]{<color>}{<title>}`
- Image is optional, colour and title are mandatory
- There are seven “official” colours: `maincolor`, `sintefdarkgreen`, `sintefgreen`, `sinteflightgreen`, `sintefred`, `sintefyellow`, `sinteflilla`.
  - Strangely enough, these are *more* than the official colours for the footline.
  - It may still be a nice touch to change the footline of following slides to the same color of a chapter slide. Your choice.



Side-Picture Slides

2 Personalization

- Opened with `\begin{sidepic}{<image>}{<title>}`
- Otherwise, `sidepic` works just like `frame`





Fonts

2 Personalization

- The paramount task of fonts is being readable
- There are good ones…
  - Use serif fonts only with high-definition projectors
  - Use sans-serif fonts otherwise (or if you simply prefer them)
- … and not so good ones:
  - `Never use monospace for normal text`
  - Gothic, calligraphic or weird fonts should always be avoided



Look

2 Personalization

- To insert a final slide with the title and final thanks, use `\backmatter`.
  - The title also appears in footlines along with the author name, you can change this text with `\footlinepayoff{...}`
  - You can remove the title from the final slide with `\backmatter[notitle]`
- The aspect ratio defaults to 16:9, and you should not change it to 4:3 for old projectors as it is inherently impossible to perfectly convert a 16:9 presentation to 4:3 one; spacings *will* break
  - The `aspectratio` argument to the `beamer` class is overridden by the SINTEF theme
  - If you *really* know what you are doing, check the package code and look for the `geometry` class.



Table of Contents

3 Colours and Backgrounds

Introduction

Personalization

Colours and Backgrounds

Summary



Changing the Colours

- Every coloured element is derived from `maincolor`
- Pick one of the named colours in the preamble with `\colorlet{maincolor}{<name>}`:
  - `unipiblue` (default), `burgundy`, `forestgreen`, `imperialpurple`, `deepteal`, `terracotta`, `slategrey`
  - Any colour defined in the `xcolor` package
- For any other colour, redefine `maincolor`; the block tint and the logos follow automatically

Code for a Custom Colour

```
\definecolor{maincolor}{RGB}{135, 200, 200}
```

maincolor

block tint



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Title and Chapter Backgrounds

- `\titlebackground{<image>}` puts a full-page image behind the title slide
- The starred version `\titlebackground*{<image>}` clips the image to a split view on the right
- Without an image, `\titlebackground*` draws the theme's natural panel (a `maincolor` panel with the white logo normally, or a white panel with the `maincolor` logo in the inverted style)
- Without an image, chapters use the opposite split background



Logos

3 Colours and Backgrounds

- Use `\dynamicshield{<color>}` to draw the shield in a given colour
- Use `\deptlogo[<width>]{<color>}` to draw the department wordmark (default width: `0.3\paperwidth`)

Code for a Custom Logo

```
\deptlogo[0.4\paperwidth]{maincolor}
\dynamicshield{white}
```



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Inverted Slides

3 Colours and Backgrounds

- The **slide style** is selected in the preamble with `\themecolor{white}` (default) or `\themecolor{main}`
- In the `main` style the background becomes `maincolor`, the text and the shield become white, and blocks and footline follow
- The title page adapts automatically: the title, subtitle, author and date turn white, and `\titlebackground*` becomes a white-panel split
- Chapter slides flip to light, remaining the opposite of the title page
- This slide is itself rendered in the `main` style and switches back to `white` afterwards; do not normally switch inside the document

Code for an Inverted Presentation

```
\themecolor{main}
\themecolor{white}
```



Table of Contents

4 Summary

Introduction

Personalization

Colours and Backgrounds

Summary



Good Luck!

4 Summary

- Enough for an introduction! You should know enough by now
- If you have corrections or suggestions, send them to me!



The LaTeX Beamer `dmpisa` theme

Thank you for listening!
Any questions?